FranchiseOps AI

Agents, a SQL + RAG Copilot and role-based access for franchise operations

Milestone 3 · Infosys Springboard Internship 7.0 · Batch 4

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Problem statement

Who

Franchise owners, outlet managers and auditors running 100+ outlets.

What goes wrong

Stock-outs, staff attrition, failed audits and bad reviews are found late, in separate tools, by people who cannot ask the data a question.

One number

Before this milestone the Copilot gave a correct, on-topic answer to only 56% of 109 test questions. Now: 94%, and it says "I do not know" instead of guessing.

Proposed solution

One question passes six layers; every number comes from SQL



Taught answers	SQL	Six agents	RAG	LLM	I do not know
50 Q&A from Admin	intents, synonyms, query shown	latest findings of each agent	BM25 + FAISS, sources [1] [2]	only rephrases the evidence	when nothing answers

Role gate on every request (JWT, 5 roles, server-side 403): Store Manager and Staff only ever see their own outlet.

Tech stack and why

FastAPI + SQLite

Small, fast, one file per database; every answer is a read-only SQL query.

JWT + bcrypt + RBAC

Role and outlet are read from the database on each request, so a lock or demotion applies at once.

RAG: BM25 + FAISS

Keywords find exact terms, embeddings find meaning; rank fusion then reranking.

Google Colab (T4)

Free GPU for the language model and translation, ngrok for a public link.

Challenges faced

Two code bases, two logins	Cause: The finished app and Milestone 2 had different sign-in and APIs.	Fix: Kept the Milestone 2 login, put its roles on every page and route, 332 tests.
Simulators did not fit the app	Cause: The team prototype used fixed inputs and no database.	Fix: Rebuilt on the database: seed, P10–P90 range, assumptions, no writes.
Copilot showed the wrong table	Cause: Weak keyword matches answered unrelated questions.	Fix: Tuned intents and synonyms, and an "I do not know" rule; measured 56% to 94%.

Results

107

seed sites crawled for PDFs (42 added)

50 / 50

questions taught and answered

56% to 94%

Copilot answer rate before and after

6

agent simulators with seed and P10–P90

332

automated tests pass

5 roles

with a demonstrated 403 refusal